

WANCHI LUCIA YEN

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PROFESSIONAL SUMMARY

AI researcher committed to building robust, reliable, and equitable language technologies for African communities. Current work spans speech translation for low-resource Bantu languages (WAKANDA AI) and Central African Sign Language recognition, where a novel transformer achieved 80.39% accuracy with a 98% smaller footprint—demonstrating that high-assurance, privacy-preserving systems can be deployed in resource-constrained environments. My mission: ensure that Deaf Cameroonians and other underserved groups are served by technology built with them, grounded in rigorous evaluation and accountable design.

EDUCATION

Carnegie Mellon University Africa

Masters of Science in Engineering Artificial Intelligence | **GPA: 3.9/4.0**

Kigali, Rwanda

2025 – 2027

Relevant Courses: Introduction to Deep Learning, Natural Language Processing, AI System Design, Data Inference and Applied Machine Learning, and AI for Engineers.

University of Buea

Bachelors of Technology in Computer Engineering | **GPA: 3.58/4.0**

Buea, Cameroon

2022 – 2025

SKILLS

- **Deep Learning:** Multimodal Transformers (MVIT V2), Neural TTS (VITS), ESPnet, ASR, End-to-End Speech Translation (AST), Bi-LSTM, CNN.
- **Efficiency:** GGUF Quantization, Model Cascading, Latency Optimization, Batch-binning, Model Compression.
- **Tools & Infrastructure:** PyTorch, Hugging Face, NVIDIA A100 Clusters, Pittsburgh Supercomputing Center (PSC), Distributed Training, MediaPipe, Git, Python (NumPy, Pandas, Scikit-learn, Matplotlib), LaTeX.
- **Research:** Low-resource Data Acquisition, Benchmarking, Technical Leadership, Multimodal Fusion, Model Robustness Evaluation, Privacy-Preserving Data Processing.
- **Languages:** English (First Language), French.

PROFESSIONAL EXPERIENCE

Vision & Language Lab, CMU Africa

Graduate Research Assistant

Kigali, Rwanda

Oct 2025 – Present

- Design and implement end-to-end speech translation pipelines for Bantu languages, focusing on reducing cascading errors, a key reliability improvement for high-stakes applications (healthcare, education).
- Build PyTorch data pipelines and distributed training scripts, with rigorous hyperparameter tuning and reproducibility checks to ensure stable model behaviour across diverse settings.

RESEARCH PROJECTS

Multimodal CASL Recognition — CASL-TransNet: SOTA 80.39% — Huggingface

- Engineered a 225-D kinematic feature matrix via **MediaPipe Holistic** and 64-frame temporal standardization, utilizing hip-centering and inter-acromion scaling to enforce scale-invariance and eliminate biometric bias.
- Implemented a **Signer-Independent** protocol using **GroupShuffleSplit** to isolate 19 unique subjects, ensuring rigorous cross-subject generalization on the CASL-W60 dataset by preventing actor-specific data leakage.
- Optimized a **PyTorch** transformer pipeline with **AMP** and Gradient Clipping, achieving full SOTA convergence in 9 minutes with a 99MB footprint, proving high-accuracy viability for African edge deployment.
- The pose-only pipeline inherently preserves privacy by discarding facial/background data, and the signer-independent evaluation provides a rigorous test of real-world robustness across new users.
- **Tech Skills:** PyTorch, MediaPipe, AMP, W&B, AdamW, GroupShuffleSplit, Kinematic Engineering, Scikit-learn, KaggleHub.

Distilling Intelligence from Noise — Under Review: Interspeech 2026 [Model]

- Engineered a synthetic data distillation pipeline utilizing **Gemini 2.5 Flash** to generate the 51-hour **Kinya-Synth-50h** corpus, providing a cost-effective solution (\$1.23/hr) to the low-resource "data desert" in Kinyarwanda.
- Fine-tuned a 1B-parameter **OuteTTS** Audio Language Model which acted as a semantic denoiser, filtering teacher artifacts to achieve a **19.2% WER** on human-held-out tests, significantly outperforming lightweight variational baselines.

- Conducted linguistic stress testing on agglutinative morphology and a blind Turing Test, demonstrating that scaling to 1-billion parameters is essential for robust distillation and high machine intelligibility in Bantu languages.
- The released models and corpus enable reproducible research and transparent evaluation; the linguistic stress testing further reveals model limitations under challenging morphological conditions – critical for safe deployment in real-world scenarios.
- **Tech Skills:** OuteTTS-1B, Gemini 2.5 Flash, Audio Language Modeling (ALM), DAC Codec, Whisper Large V3, ASR Proxy Evaluation, Kinyarwanda NLP, NVIDIA A100.

COMMUNITY ENGAGEMENT & CAPACITY BUILDING

Innovative College of Technology, University of Buea

Buea, Cameroon

Responsible AI, Data Governance, and Trustworthy Deployment Workshop Facilitator

February 2026

- Designed and delivered a hands-on workshop on privacy-preserving data processing for 25 female students, teaching conversion of RGB video to landmark arrays using MediaPipe – a privacy-by-design approach that removes identifiable visual information while preserving linguistic motion.
- Demonstrated a 98% dataset size reduction (13.9GB → 266MB) ([link](#)), enabling resource-efficient, edge-deployable AI – a critical safety consideration for real-world deployment in underserved communities.
- Emphasised data governance and sovereignty, equipping students to build AI solutions rooted in local contexts rather than importing external assumptions – a core principle of equitable and accountable AI.
- Outcome: Students successfully processed the CASL-W60 dataset and adopted a mindset focused on efficiency, privacy, and local ownership – foundational skills for developing trustworthy AI in African settings.

GS PAYSANNAT High School, Digital Inclusion Workshop

Mahama Refugee Camp, Rwanda

Python Workshop Teaching Assistant

December 2025

- Taught foundational Python programming to 56 secondary school students at Mahama Refugee Camp, introducing core concepts through interactive exercises in a resource-constrained setting.
- Provided students with their first exposure to programming, fostering equitable access to AI literacy and planting seeds for future participation in Africa's growing tech ecosystem.
- Outcome: All 56 students completed the workshop, acquiring foundational Python knowledge and building arithmetic functions and a simple calculator as their final project.

INTERESTS & ACHIEVEMENTS

- **Communication Minister**, Africa Innovators Society, CMU-Africa (December 2025)
- **1st Place Winner**, Tartan Innovate Hackathon, CMU-Africa (October 2025)
- **Mastercard Foundation Scholar Award**, CMU Africa Class of 2027
- **Top 20% Finalist**, International Youth Math Challenge (Feb 2025)
- **Certified in Cybersecurity (CC)**, ISC2 (Sep 2024)
- **SOC Tier 1 Training**, Cybersafe Foundation Graduate (Dec 2024)
- **Memberships:** Women in Tech, IEEE, Biophysical Society, ICOTEL, Open Dreams, Women In Machine Learning, Data Science Club (CMU Africa), Women In Data Science Rwanda, Zindi Africa.